

# Rana Imran Mushtaq

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## EDUCATION

**Harbin Institute of Technology (HIT), Shenzhen**, School of Physics

**Shenzhen , China**

*PhD Physics*: Condensed Matter Physics, September 2025

**Supervisor**: Dr. Deng Haojian

**Relevant Modules:**

Computational Physics, Solid State Physics, Quantum Monte Carlo Simulation, Modern Physics, Advanced Statistical Physics, Python

**Ludong University**, School of Physics and Optoelectric Engineering

**Yantai , China**

*Master of Science in Physics*: Statistical Physics, December 2024

**Thesis**: Theoretical & Numerical study on the problem of Multidimensional Anomalous Diffusion

**Supervisor**: Dr. Chunyang Wang

**Relevant Modules:**

Computational Physics, Advanced Molecular Dynamics, Monte Carlo Simulation, Modern Physics, Advanced Quantum Mechanics, Quantum Simulation, Python, MATLAB, C++

Cumulative GPA: 3.53/4.00

**University of the Punjab**, College of Physics

**Lahore, Pakistan**

*Bachelor of Science in Physics*: Solid State Physics, March 2021

**Relevant Modules:**

Statistical Mechanics, Quantum Physics, Computational Physics, Nuclear Physics, Solid State Physics, Digital Electronics, Algebra, Differential Equations, Modern Physics, Electronics Laboratory

Cumulative GPA: 3.38/4.00

## SKILLS

**Computational & Analytical**: Numerical simulations, statistical modeling, quantum Monte Carlo simulation, Langevin dynamics, anomalous diffusion, data analysis, algorithm development

**Research Methodologies**: Experimental design, machine learning, analytical solutions, data visualization, hypothesis testing, research ethics, technical writing, mathematical modeling

**Programming & Software**: Python, MATLAB, C++, Matplotlib, LaTeX, Github

## WORK EXPERIENCE

**Teaching Assistant**, TA, September 2025 – continue

**Harbin Institute of Technology (HIT) Shenzhen, China**

- Assisted in teaching university physics tutorials
- Prepared, invigilated and graded exams and assignments
- Helped students in course related issues

**Science Teacher**, Jameel Science Academy, February 2018 – February 2021

**Lahore, Pakistan**

- Developed and delivered engaging lesson plans that foster critical thinking and scientific inquiry
- Managed the computer lab, ensured a learning environment
- Collaborated with colleagues to design interdisciplinary projects that connect science with real-world applications

## PUBLICATIONS

1) **Evolution of magnetic correlation in the doped Hubbard model with altermagnetic spin splitting**, *Published*, April 28, 2026

**Journal**: Physical Review B

**Authors**: Yinlong Li, **Rana Imran Mushtaq**, Ji Liu, Wing Chi Yu, Xiaosen Yang, Cho-Tung Yip, Ho-Kin Tang

2) **Quantum Filtering and Feedback Control of Dissipative Qubits via Augmented Neural Ordinary Differential Equations**, *In Preparation*, July 15, 2025

**Journal**: Physical Review A

**Authors**: Shahid Qamar, **Mushtaq Rana Imran**, Bo Li, Ho-Kin Tang

3) **Trajectory Tracking on the Optimal Path of Two-dimensional Quadratic Barrier Escaping, *Published***, April 11, 2025

**Journal:** Chinese Physics B

**Authors:** Zengxun Zhao, Xiuying Zhang, Pengchen Zhao, Chunyang Wang, **Mushtaq Rana Imran**, Joelous Malamula N.

**DOI:** 10.1088/1674-1056/adcb99

4) **Analytical exact solution of multidimensional generalized Langevin equation, *Published***, October 04, 2024

**Journal:** Journal of Qufu Normal University

**Authors:** **Mushtaq Rana Imran**, Chunyang Wang, Shichao Zhi, Zengxun Zhao

**DOI:** 10.3969/j.issn.1001-5337.2924.4.92

5) **A Matrix Form Solution of the Multidimensional Generalized Langevin Equation in the Quadratic Potential, *Published***, August 04, 2024

**Journal:** Iranian Journal of Physics Research

**Authors:** **Mushtaq Rana Imran**, Chunyang Wang, Shichao Zhi, Zengxun Zhao, Joelous Malamula Nyasulu

**DOI:** 10.47176/ijpr.24.3.41887

6) **Theoretical and Numerical study on the fluctuation of the potential energy landscape, *Published***, May 31, 2024

**Journal:** International Journal of Modern Physics B

**Authors:** Shichao Zhi, Zengxun Zhao, **Mushtaq Rana Imran**, Pengchen Zhao, Joelous Malamula Nyasulu, Chunyang Wang

**DOI:** <https://doi.org/10.1142/s0217979225500778>

## CONFERENCES & SEMINARS

**The 1<sup>st</sup> National Youth Conference On Atomic And Molecular Physics**, August 26, 2024 **Ludong University, China**

**International Symposium on Mathematics**, Participant, May 2019 **Government College University, Lahore**

## ONLINE CERTIFICATION

**Artificial Intelligence**, Future Learn

- Gained foundation knowledge of artificial intelligence concepts, techniques, and applications

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**Backyard Meteorology, The Science of Weather**, Future Learn

- Completed a Harvard University course on weather science via edX

**Nuclear Reactor Physics**, coursera, Moscow Engineering Physics Institute MePhI

- Achieved a comprehensive understanding of nuclear reactor design, operation, and safety principles

## LEADERSHIP & VOLUNTEER EXPERIENCE

**Treasurer, Department Of Physics**, August 2017 - September 2021 **Government Science College, Lahore**

- Worked as a treasurer in Department of physics during my undergraduate degree

- Arranged Physics seminars, social activities such as plantation etc.

**Poster Presentation & Wind Turbine Model**, Participant, November 2018 **Government Science College, Lahore**

- Presented a high-efficient & low-cost wind turbine model

- Achieved an appreciation certificate on poster presentation of wind turbine model

**Global Science Magazine**, Volunteer, August 2019 **Lahore, Pakistan**

- Global Science Magazine is Pakistan's first Urdu language science magazine which promotes science in the society

- Arranged a science activity known as "Science Mela"

## LANGUAGES

**English**, CEFR Level- C1

**IELTS OVERALL BAND SCORE: 7.5**

**Chinese(Mandarin)**, Advance Level

**HSK-4 SCORE: 265/300**

**Urdu**

**Native Language**

## REFERENCE

**Dr. Chunyang Wang**, MS Supervisor

- Email: [wchy@foxmail.com](mailto:wchy@foxmail.com)

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**Dr. Deng Haojian**, PhD Supervisor

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